

BME 3303 “Introduction to Biomechanics” Syllabus (Fall 2026)

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Class Hours: 9:30-10:45AM Monday and Wednesday

Place: E202

Office hours: M and W 11AM at J337

or on Zoom by appointment

Prerequisite(s): Junior level. Statics EGE 2013 or EME 4603.

Course Content

This course provides an introduction to the mechanical principles for analyzing movements of the human body, as well as the structure and function of the musculoskeletal system. Topics include; static equilibrium, kinematics and kinetics of rigid bodies, work/energy solutions, impulse/momentum solutions, impact, power, deformation, stress-strain relationships, principal stress, strain energy, axial loading, torsion, bending, shear, combined loading and material properties. These concepts are applied to solve biomechanical problems.

Course Instructional Format

The class will have lectures, active learning modules and class discussion involving student participation and a group project. Assignments include informal reflections, homework assignments, proficiency quizzes, oral presentations, two midterm exams and a cumulative final exam. This is a three credit class consisting of two 75 min classes/week.

Required Textbook

Fundamentals of Biomechanics: Equilibrium, Motion, and Deformation 4th Edition, Ozkaya, Nordin, Goldsheyder, Leger. Springer, 2017. ISBN: 978-3319447377

Grading

- **20% Discussion Posts**
 - Formal reflections or other active learning techniques will be completed throughout the course. Discussion Board Posts will also include student interactions through Canvas.
- **10% Homework**
 - Students may discuss homework problems, but should write up solutions independently. Late submissions will not be accepted, unless previously arranged with the instructor.
- **15% Proficiency Quizzes**
 - To demonstrate proficiency in each “mastery based skill” you must correctly complete a representative analysis on a proctored quiz.
- **10% Presentations and Group Project**
 - Each student will present selected topics and answer questions from the instructor and other students. Groups will complete an engineering design project. Additional details will be provided during the semester.
- **20% Midterm Exams**
 - Two equally weighted exams will assess the material in the two sections of the course. Exams will be closed-book. No substitute exams will be offered.
- **25% Final Exam**
 - A comprehensive final exam will assess the material in whole course. Exams will be closed-book. No substitute exams will be offered.

Grading Scale

A	≥ 90%
A-	87-89.9%
B+	83-86.9%
B	80-82.9%
B-	77-79.9%
C+	73-76.9%
C	70-72.9%
C-	67-69.9%
D+	63-66.9%
D	60-62.9%
F	<60%

Attendance and Class Participation

Students should be fully prepared and engaged in class activities. This includes asking and answering questions from instructor and classmates during lectures, active learning modules, discussions and quiz reviews. Students are encouraged to discuss assignments with other students and/or the instructor, however submitted assignments should reflect the student's own work and understanding.

Violations of academic ethics will be handled according to University policies as set forth in the University's Academic Honor Code, included here by reference (see http://www.ltu.edu/currentstudents/honor_code.asp). In general, any solution obtained from any source should be properly referenced. Specific policies for assignments in this course are:

- All undergraduate students are required to write the following pledge on all academic work submitted along with a signature. If the pledge does not accompany an assignment, I can choose to not grade it.
"I have neither given nor received unauthorized aid in completing this work, nor have I presented someone else's work as my own."
- All quizzes and exams must be by individual effort. Unprogrammed calculators are allowed, but use of laptops or programmed calculators as well as observation of others' work will be considered "cheating" as defined in the Honors Code. Plagiarism will not be tolerated.

For program assessment purposes, anonymous copies of submitted student work may be kept on file.

Unforeseen circumstances may necessitate changes, additions, or deletions to this syllabus during the term.

Course Objectives:

- Demonstrate mathematical, physics, anatomy and physiology course prerequisite concepts in biomechanical applications.
- Show how biomechanics has applications in biomedical engineering and other disciplines.
- Determine the linear and angular kinematics of particles and rigid bodies for biomechanical applications.
- Determine the internal forces acting on the human body during static and dynamic activities.
- Determine the stresses and deformations in tissues subjected to various types and combinations of loading conditions.
- Determine the stress state for various orientations within a body, including the principal stresses and directions.

- Explain how the tissue properties allow the human body to function as a mechanical object.
- Describe the mechanical properties and structure of musculoskeletal tissues and explain similarities and differences between them.
- Demonstrate the ability for self-directed learning by organizing and delivering effective written and verbal communication on the biomechanical functions of the human body.
- Use the engineering design process to generate potential solutions for a biomechanical application.

Mastery Based Skills: This course implements the mastery-based learning (MBL) approach that aims to develop your engineering mindset; specifically, in the goal of persist through and learn from failure. We learn by doing, exploring, asking questions, figuring out why and how. We will make mistakes; that's a GOOD thing! Mistakes unlock new insights. We need to challenge ourselves and keep learning from every attempt and failure. This mindset will drive our course and will model an approach that will serve you well in education and as a professional.

Therefore, multiple proficiency quiz opportunities will be provided for students to demonstrate competency and develop expertise. Class time is invested in lessons, examples, demonstrations, collaborative problem solving of textbook and authentic problems and coaching. Students are expected to use out-of-class time to practice, review, learn from mistakes, collaborate with peers, explore course resources, and participate in review sessions and office hours.

BME 3303 includes five essential skillsets, each with a MBL proficiency quiz that are tied to previous physics and math courses and needed to solve statics and dynamics problems. Each quiz date is listed in the Tentative Schedule and will be closed-book written quizzes. When a student does not achieve mastery of a topic, a proficiency quiz on the same topic will be retaken at the next date. When a student has mastered the topic, they will move on to the next proficiency quiz at the next date.

1. Forces and Moments as Vectors (Chapters 1-3)
2. Free Body Diagram (Chapter 4)
3. Joint Reaction Forces/Moments (Chapter 5)
4. Projectile Motion (Chapter 7)
5. Rotational Motion about a Fixed Axis (Chapter 9)

Semester schedule (tentatively)

Class	Date	Lecture Topic	Chapter	In Class Activity	Assignment
1	8/24	LN1 Introduction	1 to 3	Video Analysis	HW 1/ Discussion Post
2	8/26	LN2 Statics Review	4		Joints Presentations
3	8/31	LN3 Linear Kinematics	6	MBL Quiz	HW 2
4	9/2	LN4 Linear Kinematics Cont.	7	Accelerometry	Discussion Post
5	9/9	LN5 Linear Kinetics	8	MBL Quiz	HW 3
6	9/14	LN6 Linear Kinetics Cont.	8	Grip Strength	Discussion Post
7	9/16	LN7 Angular Kinematics	9	MBL Quiz	HW 4
8	9/21	Group Design Project		Sport Analysis	Discussion Post
9	9/23	LN8 Angular Kinetics	10	MBL Quiz	Concept Selection
10	9/28	Group Design Project		IMU Sensors	Calibration
11	9/30	LN9 Angular Kinetics Cont.	10	MBL Quiz	HW 5
12	10/5	LN10 Work-Energy	8		Final Design
13	10/7	LN 11 Impulse-Momentum	11	MBL Quiz	
14	10/12	Exam 1 Review			Project Presentations
15	10/14			E1 Dynamics	Exam 1
16	10/19	LN12 Uniaxial Deformation	12		HW 6
17	10/21	LN13 Stress and Strain	13	Compatibility	Tissues Presentations
18	10/26	LN14 Material Properties	15		HW 7
19	10/28	LN15 Poisson's Ratio	14	Tension Testing	Discussion Post
20	11/2	LN16 Stress Transformation	14		HW 8
21	11/4	LN17 Mohr's Circle	14	Simulation	Discussion Post
22	11/9	LN18 Failure	14		HW 9
23	11/11	LN19 Torsion	14	Fractures	Discussion Post
24	11/16	LN20 Bending	14		HW 10
25	11/18	LN21 Bending Stress	14	Simulation	Discussion Post
26	11/23	LN22 Combined Loading Cont.	14		
27	11/30			E2 Deformation	Exam 2
28	12/2	Final Exam Review			
Final	TBD		Cumulative Final Exam		